NAMAA Community at AraGenre 2026: From Encoder Baselines to Self-Consistent LLM Ensembling for Hierarchical Arabic Genre Classification

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Abstract

Hierarchical, definition-guided genre classification places a short Arabic text under a broad genre and, within it, a fine-grained specific genre, using only a natural-language definition of each. We present NAMAA's submission to AraGenre 2026. Fine-tuned bi-encoder classifiers reached 0.93 hierarchical macro F1 on development via data augmentation and cross-lingual pretraining, but collapsed to as low as 0.22 on the released test set, having specialized to a hand-engineered definition style that missed the test set's larger, 74-genre taxonomy. We root-caused this gap and switched to zero-shot LLM classification, matching the best fine-tuned ensemble's test score (0.6899 hierarchical F1) with no training; layering self-consistency sampling and disagreement resolution raised this further to 0.7013, ranking 3rd of 18 teams. Specific Macro F1 (0.5114) lags well behind Broad Macro F1 (0.8913), showing that most remaining errors concentrate in fine-grained genre choices rather than broad-category mistakes. Zero-shot LLM prompting generalizes across this definition and taxonomy shift where fine-tuned, definition-conditioned encoders do not. We release our code and system resources to support reproducibility and further research.[1]

1 Introduction

Arabic text classification is complicated by rich morphology, orthographic variation, and diverse linguistic contexts. Genre classification remains underexplored relative to Arabic sentiment analysis and dialect identification. AraGenre 2026 (El-Haj et al., 2026) addresses this gap with a hierarchical benchmark spanning Modern Standard Arabic, Classical Arabic, and several dialects.

This paper addresses hierarchical Arabic genre classification at two levels: broad genres (Informative, Interactive, Creative, Religious, Legal, Learning) and specific genres (e.g., Analytical Report, Forum Discussions), reflecting how texts are characterized by purpose, style, and social context (Biber, 1989; Lee, 2001).

What distinguishes AraGenre from prior work is its definition-guided evaluation: systems interpret abstract genre definitions and apply them to unseen genre-definition combinations absent from training, favoring semantic generalization over label memorization. The evaluation set adds undiacritized text, code-switching, and multiple dialects, close to real deployment conditions.

Our system, Definition-Guided Genre Classification with Self-Consistency Resolution, combines zero-shot LLM classification with self-consistency sampling (K=3) and disagreement resolution, reaching 0.7013 Hierarchical Macro F1 on the hidden test set (versus 0.4658 for the development-set baseline) and ranking 3rd of 18 teams. Fine-tuned encoders calibrated to augmented definitions collapse from 0.93 on development to 0.22 on the hidden set once definitions diverge; zero-shot LLM classification avoids this, reaching 0.6899 F1 with no training, then improves to 0.7013 with self-consistency and disagreement resolution (Section 5); code and checkpoints are released (Section 4).

Our contributions are:

- Diagnosis of a real dev-to-test generalization collapse (0.93 to 0.22–0.44 hierarchical F1) in encoders tuned against an augmented, dev-specific definition style;
- A zero-shot, definition-guided LLM pipeline matching the best fine-tuned encoder ensemble's test score with no training data; and
- Self-consistency with disagreement resolution, lifting the final submission to 0.7013 Hierarchical Macro F1, 3rd of 18 teams.

[1]https://github.com/NAMAA-ORG/Aragenre-Definition-Guided-Genre

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2 Background

AraGenre 2026 (El-Haj et al., 2026) asks systems to classify Arabic texts at two hierarchical levels, broad and specific genre, given as a JSON object with a unique ID and text; participants submit both genres, also in JSON.

The task defines six broad genres (Informative, Interactive, Creative, Religious, Legal, Learning), each with several specific genres, and tests whether systems can generalize from limited labeled data to genre-definition configurations unseen in training.

The dataset has training, development, and hidden evaluation splits: training is intentionally limited for a low-resource setting, development supports model selection, and the hidden evaluation set holds naturally occurring Arabic texts across multiple varieties with genre-definition combinations unseen elsewhere.

The official evaluation metric is *Hierarchical Macro F1*, defined as

$$\text{Hierarchical Macro F1} = \frac{1}{2}\big(\text{Broad Macro F1} + \text{Specific Macro F1}\big). \quad (1)$$

Here, Broad Macro F1 evaluates performance over the broad genre categories, while Specific Macro F1 evaluates the fine-grained genre categories. Their average gives equal weight to both levels of the hierarchy.

The organizers provide two baselines: embedding-based (matching texts to definitions via multilingual embeddings) and retrieval-based (feeding relevant definitions to a multilingual LM), reaching DEV Hierarchical Macro F1 of 0.4658 and 0.3899 respectively.

Related work. Genre classification is a long-standing problem in computational linguistics (Biber, 1989; Lee, 2001), made difficult by wide variation in taxonomies, annotation methods, and terminology across datasets (Kuzman and Ljubešić, 2023).

Ferhat et al. (2024) adapted the Functional Text Dimensions framework for Arabic, introducing AFTD (3,400 documents, 17 categories) and evaluating traditional and neural classifiers on it; such resources address predefined categories, not unseen genre-definition interpretation.

Hierarchical Arabic classification has also been studied, via WiHArD's broad/fine-grained organization (Bouchiha et al., 2023) and a Markov-chain approach (Al-Anzi and AbuZeina, 2018); in both, the hierarchy is fixed and known in advance, unlike AraGenre, where configurations are not fully represented in training.

Kuzman and Ljubešić (2023) survey multilingual genre identification and note substantial variation in genre definitions, datasets, and annotation practices, bearing directly on AraGenre, where cross-lingual definitions supply semantic information beyond the training labels, shifting the task toward interpretation rather than memorization.

AraGenre combines aspects no single study above addresses jointly: hierarchical prediction, definition-based semantic support for limited labeled data, and generalization to genre-definition combinations unseen in training.

3 System Overview

Figure 1 summarizes our two-stage approach: fine-tuned encoder classification, then zero-shot LLM classification.

3.1 Stage 1: Fine-Tuned Encoder Classifiers

We fine-tuned multilingual sentence encoders (E5-large (Wang et al., 2024), BGE-M3 (Chen et al., 2024)) as bi-encoders scoring text against genre-definition embeddings via `MultipleNegativesRankingLoss` and cosine similarity, using TRAIN genres with augmented definitions; some variants pretrained on cross-lingual X-GENRE data (Kuzman et al., 2023). Models were selected by DEV hierarchical macro F1; several exceeded 0.90.

Diagnosing the test-set collapse. These models scored only 0.22–0.44 hierarchical F1 on TEST, whose 74-genre taxonomy included genres with no augmented-definition entry at all: they were calibrated to the augmented definitions' phrasing, not the plain official ones (one model's F1 fell from $\sim$0.85 to $\sim$0.53 under official definitions alone). Ensembling did not close the gap (Table 2); the problem was what the models matched against, not how predictions were combined.

3.2 Stage 2: Definition-Guided LLM Classification

We replaced the encoders with zero-shot classification by the DeepSeek API (`deepseek-chat`)[2] (DeepSeek-AI, 2026), prompted with the text and official definitions directly, no fine-tuning; this

[2]Routed to DeepSeek-V4 (Appendix A).

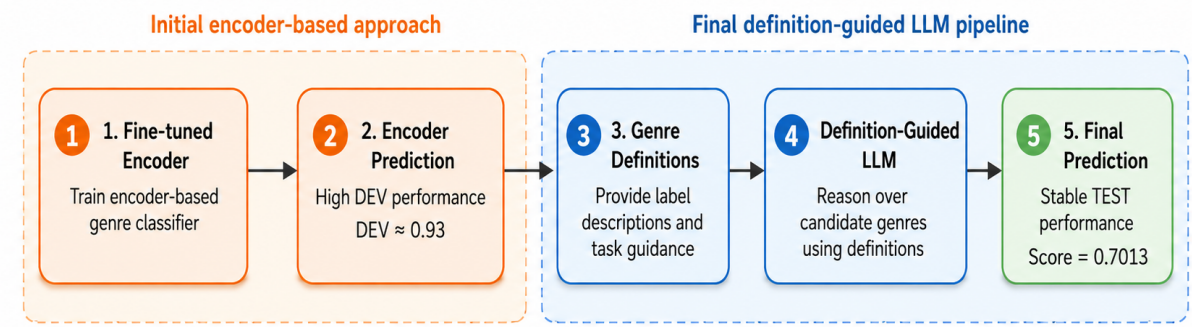


Figure 1: Pipeline of system phases

alone matched the best encoder ensemble's test score with no training data. We then applied self-consistency (Wang et al., 2023), sampling each item $K=3$ times and taking a majority vote. For items with no majority, we compared a second, reasoning-enabled DeepSeek call against a second pass by Claude (Anthropic, 2026) – both LLM-only, no human choosing or editing labels (Section 8); Claude's pass scored higher and is used in the final submission.

We also tested a dialect-guarded definition variant and a hierarchical two-stage variant (broad, then specific) using difficulty-adaptive self-consistency (Wang et al., 2025) and in-context self-verification neither outperformed disagreement resolution as shown in Table 2.

4 Experimental Setup

Data. The task provides 140 TRAIN and 110 DEV texts labeled with `broad_genre` and `specific_genre`, together with per-genre definitions; the unlabeled TEST split contains 27,972 texts covering 74 specific genres under 6 broad genres. DEV contains only 6 specific genres, one per broad genre, so its broad- and specific-genre Macro F1 are identical and not directly comparable to the much finer-grained TEST setting. The three splits share no specific-genre labels, making evaluation fully zero-shot with respect to unseen labels. Stage 1 therefore relied on genre definitions for transfer, while Stage 2 used no training data and was validated on DEV before application to TEST; some Stage 1 variants also used LLM-generated synthetic and paraphrased examples.

Self-consistency analysis. We applied self-consistency sampling with $K=3$ to all 27,972 instances in the hidden test set. The three samples agreed unanimously (3/3) on 21,018 instances (75.2%), while 6,017 instances (21.5%) yielded a majority decision (2/3). Only 937 instances (3.3%) produced three distinct predictions (1–1–1); these fully split cases were forwarded to the disagreement-resolution stage.

Evaluation and implementation. The official evaluation metric is *Hierarchical Macro F1*, computed by the CodaBench leaderboard (Xu et al., 2022) as the equal-weighted average of Broad Macro F1 and Specific Macro F1. Our encoder-based experiments were implemented with `sentence-transformers` (Reimers and Gurevych, 2019), using `intfloat/multilingual-e5-large` (Wang et al., 2024) and `BAAI/bge-m3` (Chen et al., 2024). We additionally used Hugging Face `transformers` (Wolf et al., 2020) for the cross-encoder baseline, `MoritzLaurer/mDeBERTa-v3-base-mnli-xnli` (Laurer et al., 2022), and for local inference with `Qwen/Qwen2.5-7B-Instruct` (Qwen Team, 2024; Yang et al., 2024). The definition-guided Stage 2 pipeline used the DeepSeek API with `deepseek-chat` (DeepSeek-AI, 2026).

5 Results

5.1 Development Results

Table 1 reports the full development log: roughly thirty configurations spanning 0.35 to 0.99 hierarchical F1 (Section 6 explains why this does not carry over to the hidden set).

5.2 Official Test Results

Table 2 reports official CodaBench test-set Hierarchical Macro F1 under matched procedure (no held-out or post-submission-only numbers); most of Table 1 was not submitted for test evaluation (Section 6). Our best submission, self-consistency with disagreement resolution, reached 0.7013, ranking 3rd of 18 teams (Specific Macro F1 0.5114,

System	Dev F1
Baseline & early exploratory systems	
Solution 12 – crossencoder (fp16)	0.8887
Solution 13 – Qwen local zero-shot	0.7513
Solution 14 – Qwen v2 prompted	0.8955
Solution 15 – Qwen v3 few-shot	0.9587
Solution 16 – Qwen v4 prompted	0.8736
Solution 17 – mDeBERTa cross-encoder	**0.9917**
Stage 1: single fine-tuned encoders	
S1 – BGE-M3 zero-shot	0.5828
S10 – BGE-M3 + augmented defs	0.7721
S17 (pipeline) – E5-large cosine, 8ep	0.8767
S23 – E5-large cosine, 10ep	0.8921
S23b – E5-large MNRL + X-GENRE	0.7774
S24 – E5-large MNRL + augmented defs	0.9316
S24b – E5-large MNRL + X-GENRE (ph. 1)	0.7803
S32 – E5-large + synthetic augmentation	0.3549
Multi-embedding fusion v3.0 (scientific)	0.8932
Multi-embedding fusion v3.0	0.9255
Stage 1: multi-seed runs and ensembles	
S28 – seed 42	0.9002
S28 – seed 123	0.7608
S28 – seed 777	0.8888
S28 – 3-seed average	0.9179
S29 – 8-model weighted ensemble	0.9569
S33 – 9-model weighted ensemble	0.9735
Stage 1: additional encoder/LLM screening	
S34 – mDeBERTa NLI zero-shot	0.3371
S35 – LLM zero-shot reasoning	0.8592
S36 – E5-large + paraphrase augmentation	0.7015
S37 – E5-large multi-corpus X-GENRE	0.8958
S38 – generalist ensemble, equal vote	0.9201
Stage 1: combined ensembles	
S39 – 13-model combined ensemble	**0.9917**
S40 – combined generalist, equal weight	0.9504
Stage 2: zero-shot LLM (winning path)	
DeepSeek zero-shot, no training	0.8490

Table 1: Development-set Hierarchical Macro F1; full run logs at the code repository (Section 4). Bold marks the highest score(s); Section 6 explains why this does not predict hidden-set performance.

Broad Macro F1 0.8913).

5.3 Error Analysis

Roughly a quarter of items in the hidden set lacked a unanimous self-consistency vote (21.5% partial, 3.3% split, Section 4), and only the fully split cases received disagreement resolution, leaving the partial-uncertainty items unresolved. Most errors fall within specific- rather than broad-genre classification: each specific genre maps to one broad genre, so a wrong specific label can still land on the correct broad category, consistent with Broad Macro F1 (0.8913) exceeding Specific Macro F1 (0.5114, Section 5.2). For example, one fully split case, *"wa-arja wa-qalbi kulluh jirah"* ('and I return, my heart all wounds'), resolved to `gulf_song_lyrics` after disagreeing among related song-lyric genres; another, *"sayyid al-samah hayy al-shu'ur"*, resolved to `msa_poetry` over `classical_poetry`. Both kept the correct broad genre (Creative): the disagreement is genre-internal. A two-stage broad-to-specific classification approach was tested to address this gap (Section 3) but did not outperform disagreement resolution. Definition-phrasing calibration remains difficult even for the LLM path, echoing the encoder mismatch (Section 3). The dialect-guarded variant (0.6978) also underperforms disagreement resolution (0.7013, Table 2), so explicit dialect handling did not address the dominant error source.

System	Dev F1	Test F1
Fine-tuned encoders		
S23	0.8921	0.2169
S24	0.9316	0.4415
Encoder ensembles		
Equal-vote ensemble	–	0.3666
Weighted ensemble	–	0.6899
Zero-shot LLM pipeline (winning path)		
DeepSeek zero-shot, no training	0.8490	0.6899
+ self-consistency (K=3)	–	0.6927
+ disagreement resolution	–	**0.7013**
dialect-guarded variant	–	0.6978

Table 2: Official CodaBench test-set Hierarchical Macro F1, with matched development-set scores where available (reduced 6-genre set, Section 4, not difficulty-comparable to test performance). "–" marks no comparable score under the same procedure.

6 Discussion

The development log (Table 1) is not predictive of test performance: its two highest scores (Solution 17, S39, both 0.9917) are absent from Table 2, fit to development specifics rather than generalizing; its 6 specific genres against the hidden set's 74 (Section 4) let a tightly tuned system score high without transferring. Encoders evaluated on both splits lose most of their development score there (S23: 0.8921→0.2169; S24: 0.9316→0.4415); even the best ensemble tops out at 0.6899, matched by the zero-shot LLM pipeline with no training, which improves to 0.7013 via self-consistency and disagreement resolution, degrading far less than the encoder lineage.

7 Conclusion

This work presents a definition-guided approach to hierarchical Arabic genre classification under low-resource, unseen-genre-definition conditions, reaching 0.7013 Hierarchical Macro F1 via zero shot LLM-based semantic reasoning with little labeled data. Fine-tuned encoders are more sensitive to genre-definition phrasing than direct interpretation; self-consistency with disagreement resolution adds a further, steady improvement.

8 Limitations

Gold labels for the hidden set are not public, and its taxonomy is far larger than what training and development data expose, so development-set validation does not reliably predict performance there; training data is also inherently low-resource (Section 4). Pipeline-specific: disagreement resolution covers only 3.3% of items with a fully split vote (Section 4), so most predictions get one majority-vote pass unverified; results depend on a hosted DeepSeek endpoint, and the Claude pass was run interactively rather than as an unattended batch job, though every label it produced, like every other prediction in the pipeline, came from the model, not a human.

Ethical Considerations

Genre classification errors could mislabel religious, legal, or other sensitive categories downstream; predictions should support, not replace, human review. Dialect coverage in training and development is limited relative to the hidden test set (Section 2), so performance may be uneven across underrepresented dialects.

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A Reproduction Details

Prompt template. Every LLM call uses the same system and user prompt, listing all 74 official specific-genre definitions in full and asking for one genre:

> **System:** "You are an expert Arabic text genre classifier. Identify the COMMUNICATIVE GENRE of an Arabic text – what it DOES functionally (its purpose, structure, and social role) – using only the genre definitions provided to you. Always respond with valid JSON only. No markdown."
>
> **User:** the name, broad genre, and definition text of each of the 74 genres, then the input text, then: "Read the text and the genre definitions above. Reason briefly about which definition's communicative purpose and structure best matches the text, then choose exactly one genre. Genre MUST be one of: [list]. Respond ONLY with valid JSON: {"reasoning": "1 short sentence", "genre": "<exact_genre_name>"}"

Sampling settings. Zero-shot pass (Table 2, 0.6899): DeepSeek `deepseek-chat`, temperature 0, `max_tokens`= 120, one call per item. Self-consistency pass: same prompt, K=3 samples per item at temperature 0.6 (DeepSeek `deepseek-v4-flash`, the model id backing `deepseek-chat` after its alias was deprecated, with `thinking` disabled), majority vote over the 3 sampled genre labels. Disagreement resolution: the 937 items with a fully split (1–1–1) vote were classified by Claude, given each text and all 74 definitions directly inside an interactive Claude Code session rather than a scripted batch call; every genre label came from the model's own output, with no human selecting or editing a prediction. This ran after a DeepSeek reasoning-enabled tiebreak call (`thinking` enabled) scored lower on the same slice and was not used in the final submission. Full prompt-construction and sampling code is at the repository linked in the abstract.